

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Industrial and Systems Engineering

2. Subject Name: Statistical Learning Lab
L-T-P: 0-0-3 Credits: 2

3. Pre-requisites:

Operations Research I (IM21201)
Operations Research II (IM21204)
Probability and Statistics (MA20205)

4. Syllabus and reference books:

Syllabus:

Data Preprocessing, Handling and Visualization, Linear Regression, Classification, Resampling, Model Selection and Regularization, Non-linear Regression, Tree-based Methods, Artificial Neural Network, Support Vector Machines, Unsupervised Learning

Text Books:

1) James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning, New York: Springer.

Reference Books:

1) Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: data mining, inference, and prediction. Springer Science & Business Media.
2) Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). Mining of massive data sets. Cambridge university press.
3) Haykin, S. S. (2009). Neural networks and learning machines/Simon Haykin.
4) Bishop, C. M. (2006). Pattern Recognition and Machine learning. Springer.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of Sessions
1.	Data Preprocessing, Handling and Visualization	1
2.	Linear Regression	1
3.	Classification	1
4.	Resampling Methods	1
5.	Model Selection and Regularization	1
6.	Non-linear Regression	1
7.	Tree-Based Methods	1

8.	Artificial Neural Networks	2
9.	Support Vector Machines	1
10.	Unsupervised Learning	2
Total number of sessions		12